MACHAKOS UNIVERSITY
CIT department
Lecture 2 & 3

By: Ben. Kiage

Data Mining as the Evolution of Information Technology

Data mining can be viewed as a result of the natural evolution of information technology.
The database and data management industry evolved in the development of several critical functionalities:

- ✓ *data collection and database creation,*
- ✓ *data management* (including data storage and retrieval and database transaction processing), and
- ✓ *Advanced data analysis* (involving data warehousing and data mining).

The early development of data collection and database creation mechanisms served as a prerequisite for the later development of effective mechanisms for data storage and retrieval, as well as query and transaction processing.

Nowadays numerous database systems offer query and transaction processing as common practice. Advanced data analysis has naturally become the next step.

Since the 1960s, database and information technology has evolved systematically from primitive file processing systems to sophisticated and powerful database systems.

The research and development in database systems since the 1970s progressed from early hierarchical and network database systems to relational database systems (where data are stored in relational table structures, data modeling tools, and indexing and accessing methods.
In addition, users gained convenient and flexible data access through query languages, user interfaces, query optimization, and transaction management.

Efficient methods for online transaction processing (OLTP), where a query is viewed as a read-only transaction, contributed substantially to the evolution and wide acceptance of relational technology as a major tool for efficient storage, retrieval, and management of large amounts of data.

After the establishment of database management systems, database technology moved toward the development of advanced database systems, data warehousing, and data mining for advanced data analysis and web-based databases.

Advanced database systems, for example, resulted from an upsurge of research from the mid-1980s onward.

These systems incorporate new and powerful data models such as extended-relational, object-oriented, object-relational, and deductive models.

Application-oriented database systems have flourished, including spatial, temporal, multimedia, active, stream and sensor, scientific and engineering databases, knowledge bases, and office information bases.

Issues related to the distribution, diversification, and sharing of data have been studied extensively.

Advanced data analysis sprang up from the late 1980s onward. The steady and dazzling progress of computer hardware technology in the past three decades led to large supplies of powerful and affordable computers, data collection equipment, and storage media.

This technology provides a great boost to the database and information industry, and it enables a huge number of databases and information repositories to be available for transaction management, information retrieval, and data analysis.

Data can now be stored in many different kinds of databases and information repositories. One of the emerging data repository architecture is the data warehouse.

This is a repository of multiple heterogeneous data sources organized under a unified Schema at a single site to facilitate management decision making.

Data warehouse technology includes data cleaning, data integration, and online analytical processing (OLAP)—that is, analysis techniques with functionalities such as summarization, consolidation, and aggregation, as well as the ability to view information from different angles.

Although OLAP tools support multidimensional analysis and decision making, additional data analysis tools are required for in-depth analysis

For example, data mining tools that provides data classification, clustering, outlier/anomaly detection, and the characterization of changes in data over time.

Huge volumes of data have been accumulated beyond databases and data warehouses.

During the 1990s, the World Wide Web and web-based databases (e.g., XML databases) began to appear.

Internet-based global information bases, such as the WWW and various kinds of interconnected, heterogeneous databases, have emerged and play a vital role in the information industry.

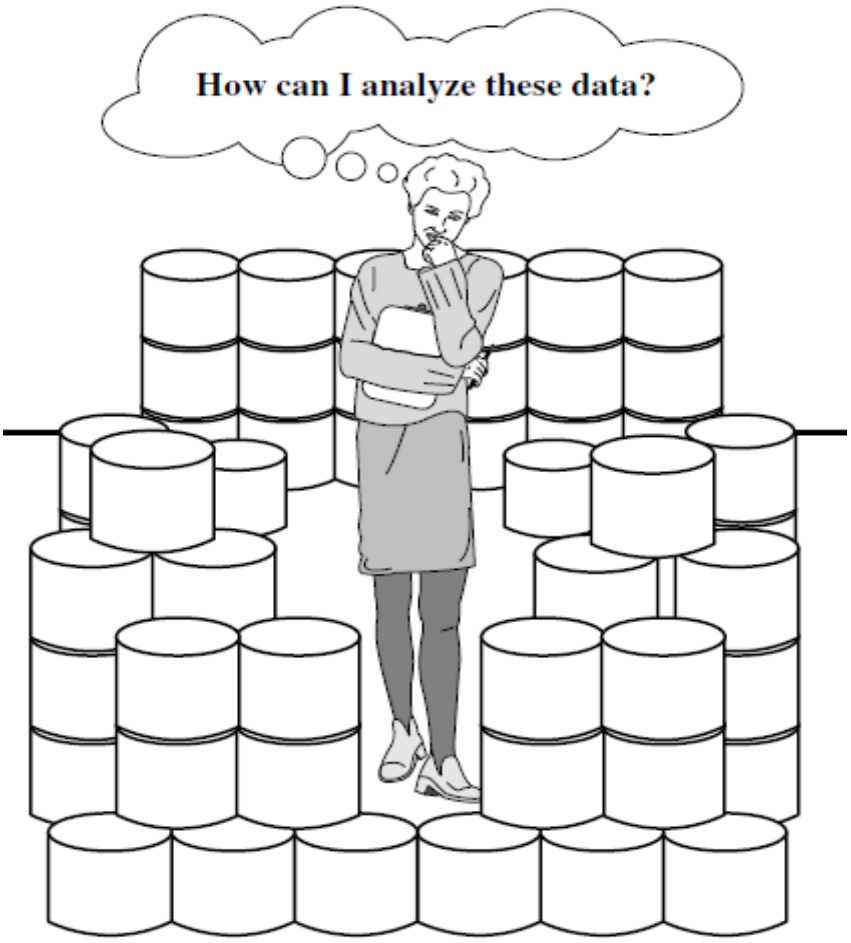
The effective and efficient analysis of data from such different forms of data by integration of information retrieval, data mining, and information network analysis technologies is a challenging task.

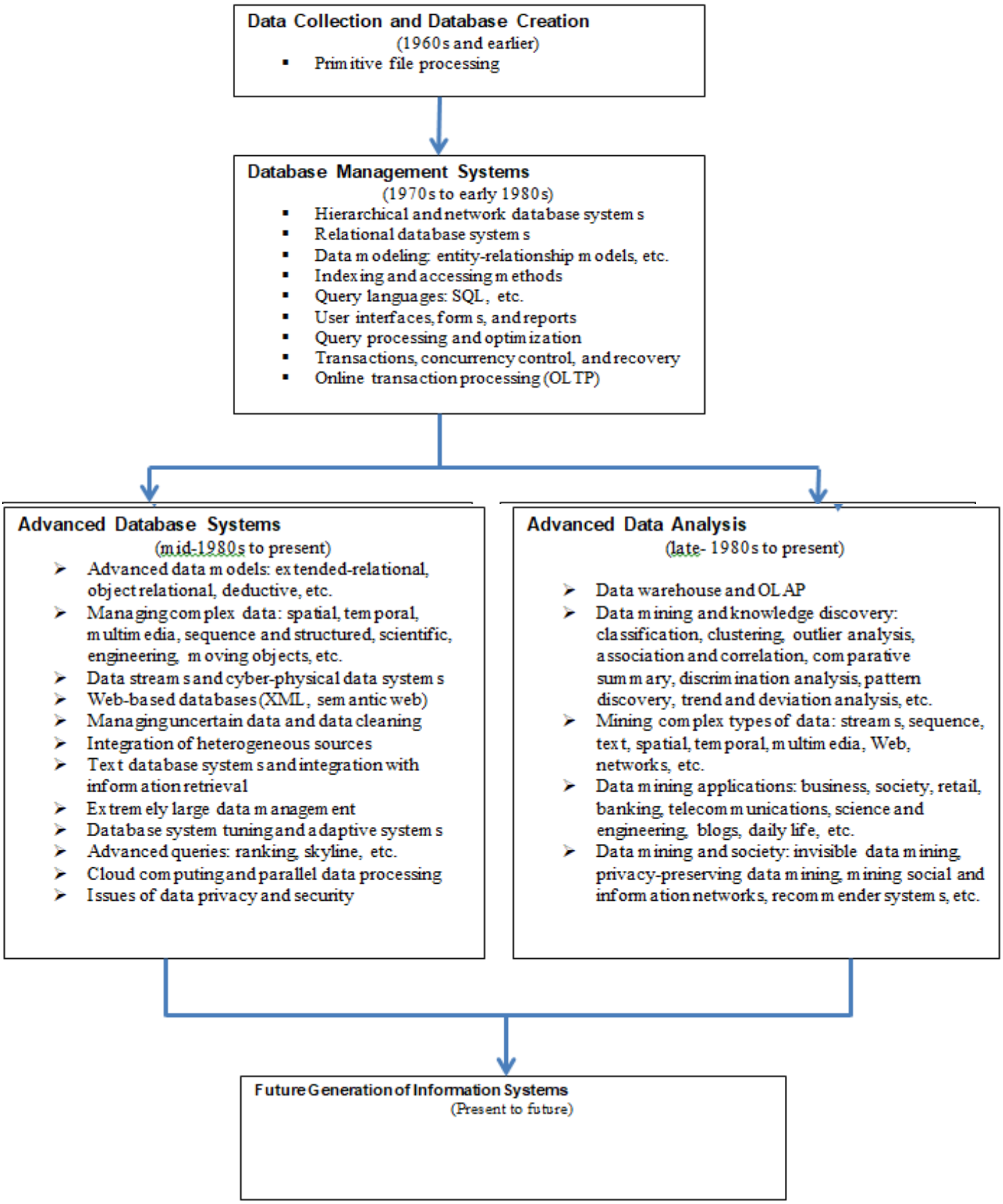
NOTE

Efforts have been made to develop expert system and knowledge-based technologies, which typically rely on users or domain experts to manually, input knowledge into knowledge bases.

Unfortunately, however, the manual knowledge input procedure is prone to biases and errors and is extremely costly and time consuming.

The widening gap between data and information calls for the systematic development of data mining tools that can turn data tombs into “golden nuggets” of knowledge.



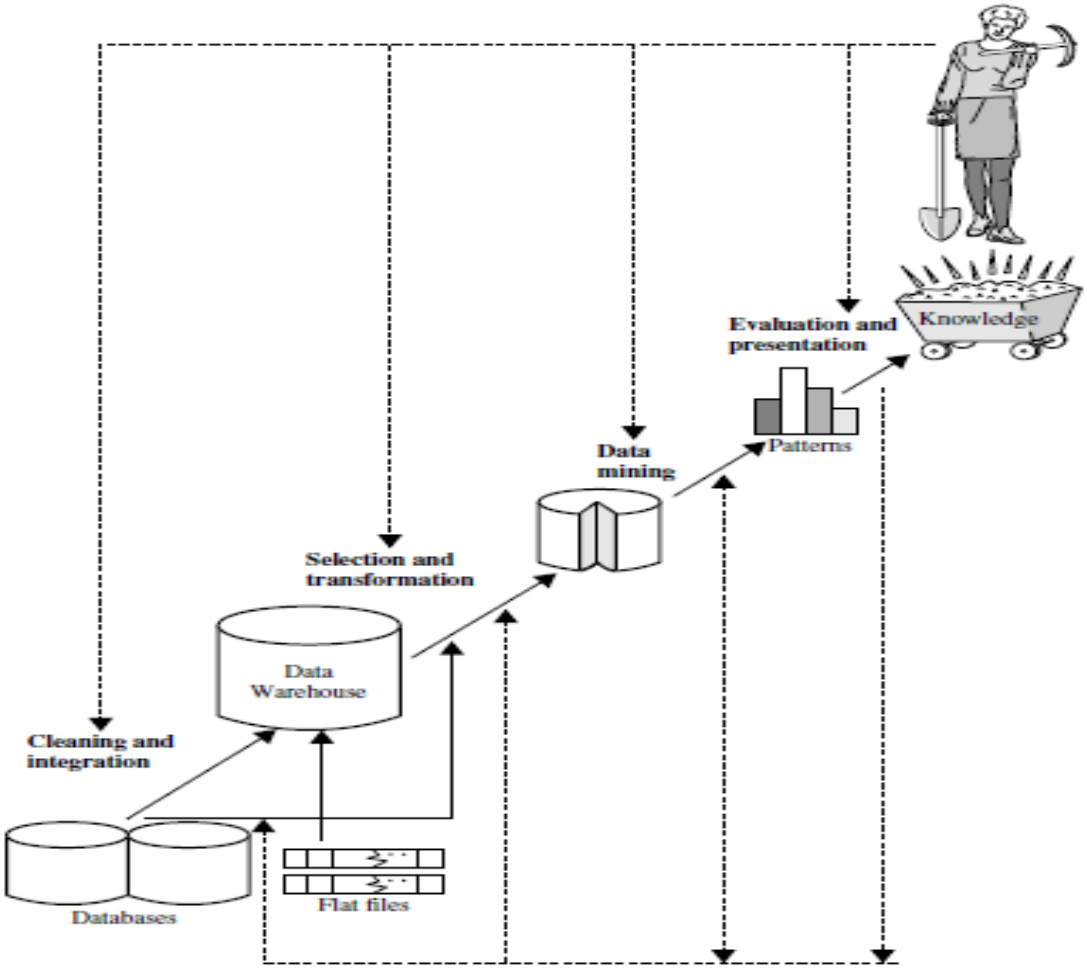


The evolution of database system technology

Data mining is the *process* of discovering interesting patterns and knowledge from *large* amounts of data. The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.

The knowledge discovery process is shown in Figure below as an iterative sequence of the following steps:

1. **Data cleaning** (to remove noise and inconsistent data)
2. **Data integration** (where multiple data sources may be combined)
3. **Data selection** (where data relevant to the analysis task are retrieved from the database)
4. **Data transformation** (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)
5. **Data mining** (an essential process where intelligent methods are applied to extract data patterns)
6. **Pattern evaluation** (to identify the truly interesting patterns representing knowledge based on *interestingness measures*)
7. **Knowledge presentation** (where visualization and knowledge representation techniques are used to present mined knowledge to users)



Steps 1 through 4 are different forms of data preprocessing, where data are prepared for mining. The data mining step may interact with the user or a knowledge base.

The interesting patterns are presented to the user and may be stored as new knowledge in the knowledge base.

Data Warehouses

A **data warehouse** is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site.

Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.

Suppose that AllElectronics is a successful international company with branches around the world. Each branch has its own set of databases.

The president of AllElectronics has asked you to provide an analysis of the company's sales per item type per branch for the third quarter.

This is a difficult task, particularly since the relevant data are spread out over several databases physically located at numerous sites.

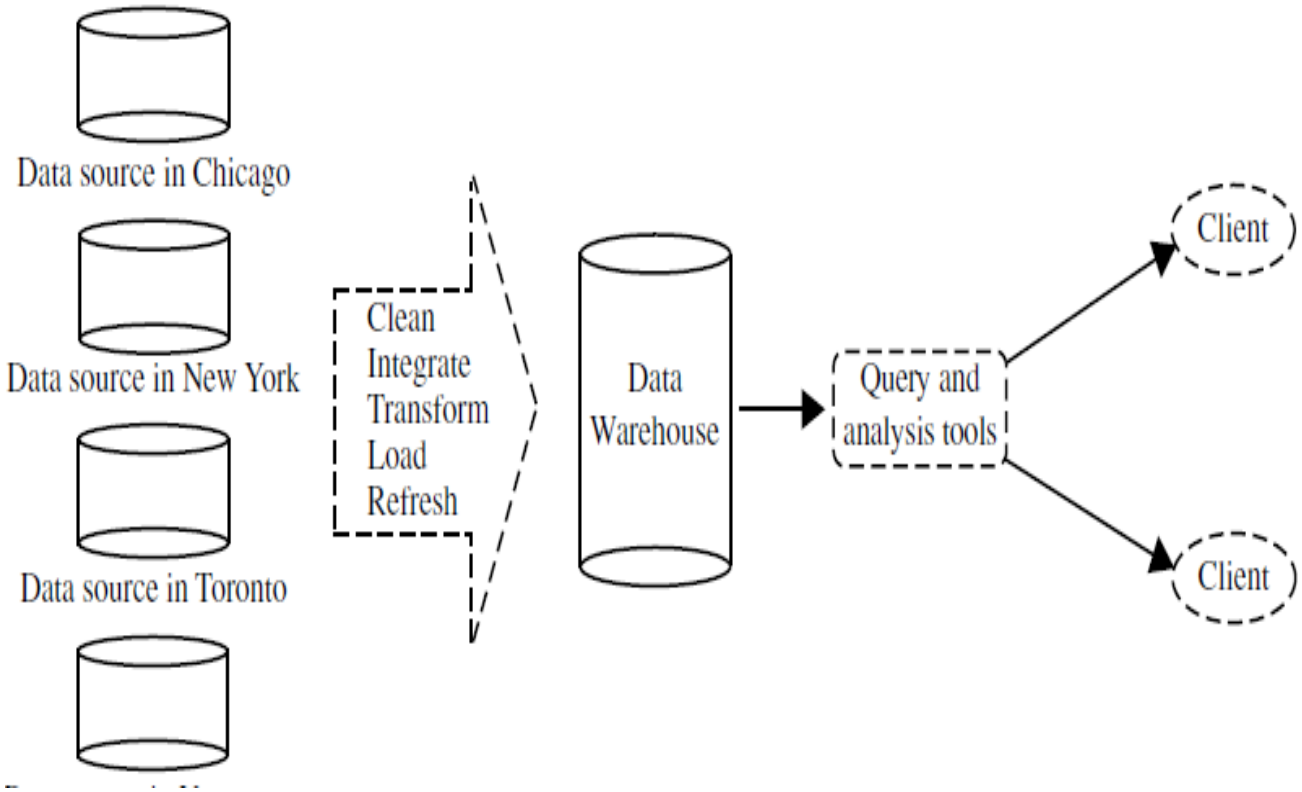
To facilitate decision making, the data in a data warehouse are organized around major subjects (e.g., customer, item, supplier, and activity).

The data are stored to provide information from a historical perspective, such as in the past 6 to 12 months, and are typically summarized.

For example, rather than storing the details of each sales transaction, the data warehouse may store a summary of the transactions per item type for each store or, summarized to a higher level, for each sales region.

A data warehouse is usually modeled by a multidimensional data structure, called a data cube, in which each dimension corresponds to an attribute or a set of attributes in the schema, and each cell stores the value of some aggregate measure such as count or sum.

A data cube provides a multidimensional view of data and allows the pre-computation and fast access of summarized data.



A data cube for AllElectronics.
A data cube for summarized sales data of AllElectronics is presented in Figure below. The cube has three dimensions:

Address (with city values
Chicago, New York, Toronto, Vancouver),

Time (with quarter values Q1, Q2, Q3, Q4), and

Item (with item type values home entertainment, computer, phone, security)

The aggregate value stored in each cell of the cube is sales amount (in thousands).

For example, the total sales for the first quarter, Q1, for the items related to security systems in Vancouver is $400, 000, as stored in cell Vancouver, Q1, security.

Additional cubes may be used to store aggregate sums over each dimension, corresponding to the aggregate values obtained using different SQL group-by (e.g., the total sales amount per city and quarter, or per city and item, or per quarter and item, or per each individual dimension).

By providing multidimensional data views and the pre-computation of summarized data; data warehouse systems can provide inherent support for OLAP. Online analytical processing operations make use of background knowledge regarding the domain of the data being studied to allow the presentation of data at different levels of abstraction.

Such operations accommodate different user viewpoints. Examples of OLAP operations include drill-down and roll-up, which allow the user to view the data at differing degrees of summarization, For instance, we can drill down on sales data summarized by quarter to see data summarized by month.

Similarly, we can roll up on sales data summarized by city to view data summarized by country.

Although data warehouse tools help support data analysis, additional tools for data mining are often needed for in-depth analysis.

Multidimensional data mining (also called **exploratory multidimensional data mining**) performs data mining in multidimensional space in an OLAP style. That is, it allows the exploration of multiple combinations of dimensions at varying levels of granularity in data mining, and thus has greater potential for discovering interesting patterns representing knowledge.

Data mining functionalities, and the kinds of patterns they can discover, are described below. Interesting patterns represent knowledge.

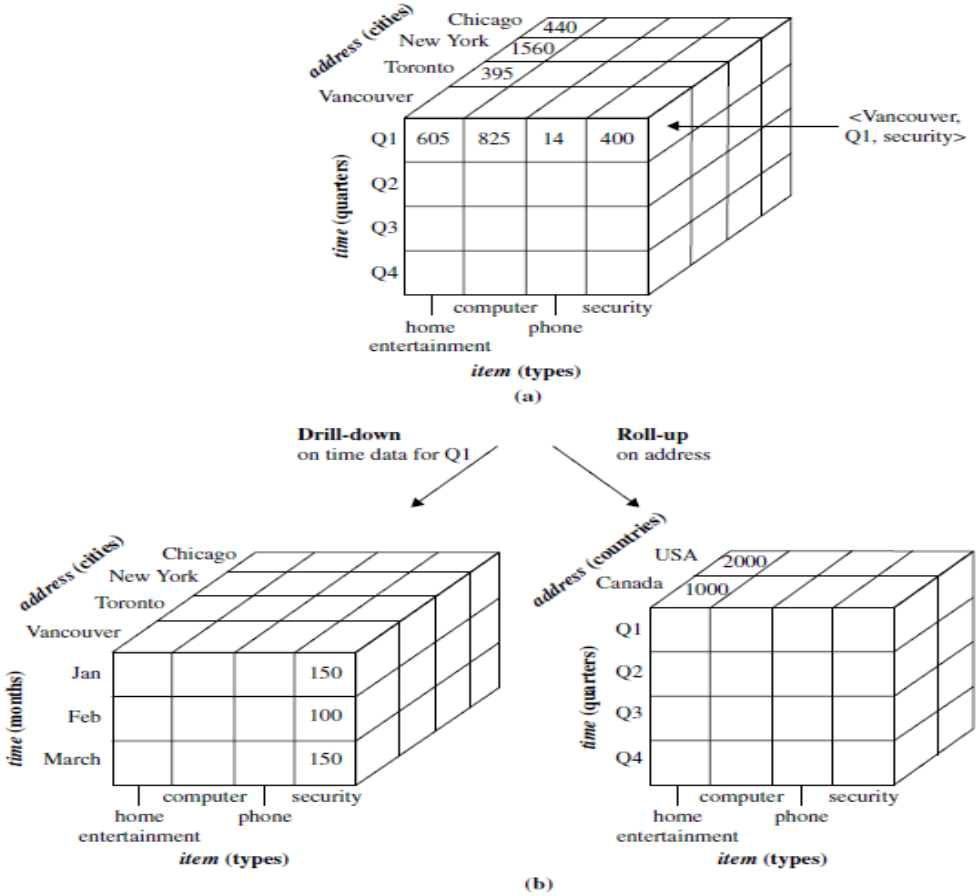


Figure: multidimensional data cube, commonly used for data warehousing, (a) showing summarized data for *AllElectronics* and (b) showing summarized data resulting from drill-down and roll-up operations on the cube in (a). For improved readability, only some of the cube cell values are shown.

What Kinds of Patterns Can Be Mined?

Having observed various types of data and information repositories on which data mining can be performed. Let us now examine the kinds of patterns that can be mined.

There are a number of data mining functionalities. These include characterization and discrimination; the mining of frequent patterns, associations, and correlations; classification and regression; clustering analysis; and outlier analysis.

Data mining functionalities are used to specify the kinds of patterns to be found in data mining tasks.

In general, such tasks can be classified into two categories: descriptive and predictive.

Descriptive mining tasks characterize properties of the data in a target data set.

Predictive mining tasks perform induction on the current data in order to make predictions.

Class/Concept Description: Characterization and Discrimination

Data entries can be associated with classes or concepts.
For example, in the *AllElectronics* store, classes of items for sale include *computers* and *printers*, and concepts of customers include *big Spenders* and *budget Spenders*.

It can be useful to describe individual classes and concepts in summarized, concise, and yet precise terms. Such descriptions of a class or a concept are called **class/concept descriptions**.
These descriptions can be derived using

1) ***Data characterization***, by summarizing the data of the class under study (often called the **target class**) in general terms, or

2) ***Data discrimination***, by comparison of the target class with one or a set of comparative classes (often called the **contrasting classes**), or

3) Both data **characterization** and **discrimination**.

Data characterization
Data characterization is a summarization of the general characteristics or features of a target class of data. The data corresponding to the user-specified class are typically collected by a query.

For example, to study the characteristics of software products with sales that increased by 10% in the previous year, the data related to such products can be collected by executing an SQL query on the sales database.

There are several methods for effective data summarization and characterization. Including Simple data summaries based on statistical measures and plots.

The data cube-based OLAP roll-up operation can be used to perform user-controlled data summarization along a specified dimension.

An *attribute-oriented induction* technique can be used to perform data generalization and characterization without step-by-step user interaction. The output of data characterization can be presented in various forms.

Examples include **pie charts**, **bar charts**, **curves**, **multidimensional data cubes**, and **multidimensional tables**, including crosstabs. The resulting descriptions can also be presented as **generalized relations** or in rule form (called **characteristic rules**).

Example: Data characterization
A customer relationship manager at *AllElectronics* may order the following data mining task: ……
Summarize the characteristics of customers who spend more than $5000 a year at AllElectronics. The result is a general profile of these customers, such as that they are 40 to 50 years old, employed, and have excellent credit ratings. The data mining system should allow the customer relationship manager to drill down on any dimension, such as on *occupation* to view these customers according to their type of employment.

Data discrimination

Data discrimination is a comparison of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.

The target and contrasting classes can be specified by a user, and the corresponding data objects can be retrieved through database queries.

For example, a user may want to compare the general features of software products with sales that increased by 10% last year against those with sales that decreased by at least 30% during the same period. The methods used for data discrimination are similar to those used for data characterization.

"How are discrimination descriptions output?"
The forms of output presentation are similar to those for characteristic descriptions, although discrimination descriptions should include comparative measures that help to distinguish between the target and contrasting classes.
Discrimination descriptions expressed in the form of rules are referred to as **discriminant rules**.

Example: Data discrimination

A customer relationship manager at *AllElectronics* may want to compare two groups of customers—those who shop for computer products regularly (e.g., more than twice a month) and those who rarely shop for such products (e.g., less than three times a year).

The resulting description provides a general comparative profile of these customers, such as that 80% of the customers who frequently purchase computer products are between 20 and 40 years old and have a university education, whereas 60% of the customers who infrequently buy such products are either seniors or youths, and have no university degree.

Drilling down on a dimension like *occupation*, or adding a new dimension like *income level*, may help to find even more discriminative features between the two classes.

Mining Frequent Patterns, Associations, and Correlations

Frequent patterns

These, are patterns that occur frequently in data.

There are many kinds of frequent patterns, including frequent itemsets, frequent subsequences (also known as sequential patterns), and frequent substructures.

A *frequent itemset typically* refers to a set of items that often appear together in a transactional data set—for example, milk and bread, which are frequently bought together in grocery stores by many customers.

A frequently occurring subsequence, such as the pattern that customers, tend to purchase first a laptop, followed by a digital camera, and then a memory card, is a (*frequent*) *sequential pattern*.

A substructure can refer to different structural forms (e.g., graphs, trees, or lattices) that may be combined with itemsets or subsequences.

If a substructure occurs frequently, it is called a (*frequent*) *structured pattern*. Mining frequent patterns leads to the discovery of interesting associations and correlations within data.

Example: Association analysis

Suppose that, as a marketing manager at *AllElectronics*, you want to know which items are frequently purchased together (i.e., within the same transaction).

An example of such a rule, mined from the *AllElectronics* transactional database, is

Buys(*X*, "*computer*") ⇒ *buys*(*X*, "*software*") [*support* =1%, *confidence* = 50%],

Where *X* is a variable representing a customer,

A **confidence**, or certainty, of 50% means that if a customer buys a computer, there is a 50% chance that she will buy software as well.

A 1% **support** means that 1% of all the transactions under analysis show that computer and software are purchased together. This association rule involves a single attribute or predicate (i.e., *buys*) that repeats.

Association rules that contain a single predicate are referred to as **single-dimensional association rules**. Dropping the predicate notation, the rule can be written simply as "*computer* ⟹ *software* [1%, 50%]."

Suppose, instead, that we are given the *AllElectronics* relational database related to purchases.
A data mining system may find association rules like:-

Age(*X*, "20...29") ^*income*(*X*, "40K...49K") *buys*(*X*, "laptop")

[*Support* = 2%, *confidence* = 60%].

The rule indicates that of the *AllElectronics* customers under study, 2% are 20 to 29 years old with an income of $40,000 to $49,000 and have purchased a laptop (computer) at *AllElectronics*.

There is a 60% probability that a customer in this age and income group will purchase a laptop.

Note that this is an association involving more than one attribute or predicate (i.e., *age, income,* and *buys*).

Adopting the terminology used in multidimensional databases, where each attribute is referred to as a dimension, the above rule can be referred to as a **multidimensional association rule**.

Typically, association rules are discarded as uninteresting if they do not satisfy both a **minimum support threshold** and a **minimum confidence threshold**.

Additional analysis can be performed to uncover interesting statistical **correlations** between associated attribute–value pairs.

Frequent itemset mining is a fundamental form of frequent pattern mining. The mining of frequent patterns, associations, and correlations is where particular emphasis is placed on efficient algorithms for frequent itemset mining.

DRM Software Reviews